

MACEIO, Brazil

WMO: 829930

Lat: 9.511S

Lon: 35.792W

Elev: 387

StdP: 14.49

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	65.3	66.3	62.3	85.1	74.6	63.5	89.1	74.1	19.1	78.0	17.8	78.2	3.3	320	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	14.2	91.5	77.9	89.9	77.2	89.3	77.0	80.3	86.5	79.6	85.6	79.0	85.0	12.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	152.5	83.2	78.1	148.3	82.5	77.3	144.0	81.7	44.4	86.7	43.5	85.9	42.7	85.4	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.2	16.5	15.3	DB	63.0	96.0	1.8	2.2	61.7	97.6	60.7	98.9	59.7	100.1	58.4	101.7
			WB	61.8	82.8	2.1	1.9	60.3	84.2	59.1	85.3	58.0	86.3	56.5	87.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.2	80.3	80.7	81.1	80.0	77.8	76.0	74.6	74.5	75.7	77.7	79.3	80.3
	DBStd	3.07	2.28	2.11	2.01	1.85	1.97	2.01	2.11	1.95	1.90	1.85	1.98	1.89
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10275	939	860	965	899	863	779	764	760	772	857	878	939
	CDD65	4800	474	440	500	449	398	329	299	295	322	392	428	474
	CDH74	37018	4290	4118	4764	3834	2719	1760	1340	1458	2012	2890	3596	4237
	CDH80	11488	1548	1496	1775	1243	664	273	141	181	401	866	1308	1592
Wind	WSAvg	6.8	7.4	6.9	6.4	5.8	5.6	5.9	6.2	6.4	7.0	7.6	8.0	8.0
Precipitation	PrecAvg	58.9	2.3	3.0	4.6	7.2	9.1	10.2	8.5	5.5	3.8	2.5	1.1	1.2
	PrecMax	79.7	7.9	9.4	11.4	19.1	23.5	19.4	16.2	11.8	12.1	9.3	5.8	3.7
	PrecMin	32.9	0.0	0.4	1.4	1.8	1.7	3.4	2.9	1.7	0.6	0.0	0.0	0.1
	PrecStd	12.6	2.0	2.0	2.3	3.7	4.5	4.4	3.1	2.3	2.9	2.3	1.4	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	92.1	91.8	91.4	88.2	85.9	84.5	85.6	87.3	89.4	91.5	93.0
		MCWB	78.9	78.4	78.2	78.0	77.2	76.5	75.8	74.9	76.7	75.6	77.7	77.9
	2%	DB	90.1	90.0	90.4	89.4	86.4	84.2	82.7	83.1	84.8	87.6	89.5	90.0
		MCWB	77.5	77.5	77.6	77.4	76.7	75.6	74.6	73.9	74.7	75.7	76.6	76.9
	5%	DB	88.2	89.2	89.5	87.7	85.4	82.7	81.8	82.2	84.0	86.1	87.9	88.7
		MCWB	76.2	77.1	77.4	77.2	76.3	74.7	74.0	73.3	74.2	75.0	75.7	76.3
	10%	DB	87.3	87.6	87.9	86.2	84.0	81.9	80.4	80.7	82.4	84.5	86.3	87.5
		MCWB	75.9	76.5	76.8	76.9	75.9	74.3	73.1	72.6	73.4	74.4	75.2	75.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	80.7	81.4	80.9	80.1	78.9	78.5	77.7	78.3	78.8	79.8	80.3
		MCDB	88.4	87.0	87.8	85.8	84.5	82.7	81.7	81.2	83.9	85.0	87.2	87.8
	2%	WB	79.6	79.6	79.9	79.7	78.9	77.4	77.0	76.3	76.9	77.4	78.5	79.0
		MCDB	86.0	85.9	86.2	85.1	83.6	81.5	80.5	80.3	82.1	83.8	85.7	86.3
	5%	WB	78.7	78.8	79.1	78.9	78.0	76.5	75.9	75.3	75.9	76.5	77.6	78.1
		MCDB	84.8	85.1	85.3	84.4	82.6	80.5	79.3	79.5	81.0	82.8	84.5	85.1
	10%	WB	77.9	78.1	78.5	78.2	77.0	75.7	74.7	74.3	74.9	75.7	76.6	77.3
		MCDB	83.8	84.2	84.5	83.5	81.5	79.5	78.1	78.4	79.9	81.9	83.2	83.9
Mean Daily Temperature Range	5% DB	MDBR	14.4	14.2	14.2	13.1	12.3	11.7	11.7	13.0	14.1	15.0	15.7	15.4
		MCDBR	15.9	15.9	15.8	15.0	14.2	13.0	13.2	14.7	15.9	16.6	17.1	16.9
		MCWBR	5.5	5.3	5.4	5.6	6.3	6.3	6.7	7.1	7.2	6.6	6.1	5.7
	5% WB	MCDBR	14.6	14.2	14.0	13.1	12.2	11.4	11.5	12.7	14.1	15.1	15.5	15.5
		MCWBR	6.3	5.9	5.9	6.2	6.5	6.4	7.1	7.6	7.6	7.0	6.7	6.5
Clear-Sky Solar Irradiance	taub	0.386	0.386	0.386	0.378	0.374	0.373	0.371	0.382	0.410	0.408	0.393	0.396	
	taud	2.510	2.519	2.524	2.560	2.577	2.563	2.556	2.489	2.385	2.403	2.458	2.462	
	Ebn at Noon	303	301	296	289	281	277	280	284	285	291	298	299	
	Edh at Noon	36	36	35	32	30	30	30	34	39	40	38	38	
All-Sky Solar Radiation	RadAvg	2032	2002	1970	1793	1582	1447	1514	1685	1886	2012	2154	2125	
	RadStd	153	84	85	108	126	93	85	86	96	87	100	82	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	-0.74	-1.21	-2.07	N/A	-1.22	-1.33	N/A	N/A	-281	-279
Regional (1 neighbor)	-1.15	-0.94	-1.67	-1.42	-1.19	-1.08	N/A	N/A	-475	-479

Nomenclature: See separate page